

Scenario-Robust Traffic State Reconstruction and Queue Forecasting

IEEE BigData Cup 2026 — TrafficFlowBench · Team Steins · final submissions V14a (public 0.88147) and V13 (0.88107)

1. Summary

Validation (March) and private (April) are **new traffic scenarios**: the scenario seed is redrawn, so per-link free-flow speeds, queued-speed levels, ramp demand and the bottlenecks that activate all change. A model that is validated only on held-out training days cannot see this shift. Our solution is built around one discipline: **every change is accepted only on out-of-scenario evidence**, measured on the published validation/private layers (for Task 1, which is offline, the final model also adapts to them; Task 2 never sees data after its origin). With that discipline in place, four structural findings carried most of the gain:

1. **Measurement noise has a corridor-wide common mode.** The free-flow branch of the generator's fundamental diagram is flat, so a free-flowing speed minus its link plateau is pure measurement noise. That noise shares a component across the whole corridor at each 5-minute slot, and the other links observed at the same slot estimate it.
2. **Scenario statistics must come from the scenario itself.** Plateaus and profiles measured on the training months plant a per-link bias in the new months. Measured on the split's own published layer (Task 1 is offline, so this is allowed), they remove about 0.3 km/h of speed RMSE out of scenario.
3. **Published layers carry sensors that were overlooked.** The masked layer publishes the Task 2 origin row T at about 58% coverage. And on-ramp flow collapses while the mainline link it feeds is queued, while the ramp layer stays published inside the 90-minute mainline blackouts that follow each Task 2 origin.
4. **Task 1 can learn the new scenario from its own published cells.** Task 1 targets are Bernoulli-masked cells, so hiding further observed cells of the validation/private layer gives in-scenario training rows with the same structure. A correction booster fitted on them (test-time adaptation, frozen backbone) beats the train-only model on 19/20 panel-splits.

From the inherited pipeline (public 0.87446) the final submission reaches **0.88147**. Our out-of-scenario Task 1 evaluator predicted the last three Task 1 deltas to within 0.0005; Section 6.3 reports the one change it could not judge, and what it cost.

2. Tasks, rules and validation

Score. $S = 0.35 \cdot S_{\text{state}} + 0.30 \cdot S_{\text{queue}} + 0.15 \cdot S_{\text{physics}} + 0.20 \cdot S_{\text{ODME}}$. Task 3 is computed from the Task 1 answer.

Information sets. We follow the organizers' ruling (forum topic 742068). Task 2 is online: a forecast may only use released data with a timestamp at or before the origin T, in any file. Task 1 is offline: any released observation of the split may be used. Our Task 2 inputs are the 60-minute window history, the masked-layer origin row at T, and the same-day and earlier-day masked layer before T. Ramp values inside the horizon and mainline observations after the post-horizon buffer are never read for Task 2.

Out-of-scenario evaluators.

- *Task 1, hidden cells.* We hide 3% of the observed eligible cells of validation/private (about 39k per panel-split) and rebuild every feature without them. This includes the split-own plateaus and profiles, so the hidden values never enter their own predictors. We then score the

predictions against the published values (tfb/t1_shift_eval.py). Synthetic blackouts (all links blank for 18 slots) score the gap specialist the same way.

- **Task 2, mined events.** We take onset events and ongoing windows found in the validation/private masked layer and build their inputs exactly as at inference (tfb/t2_onset_shift.py, tfb/t2_prod_shift.py). Ongoing windows use the organizers' selector rule (persistence IoU ≤ 0.9).

Adoption rule. A paired gain of more than 2 standard errors. Both splits must be non-negative. At least half of the changed panels must improve in both splits. The gain must survive dropping the best panel.

Change	Predicted Δ	Online Δ	Evaluator
V10a $\rightarrow$ V12	+0.004 ... +0.0055	-0.0012	train-day holdout (blind to the scenario shift)
V12 $\rightarrow$ V12b	+0.006	+0.0058	out-of-scenario hidden cells
V12b $\rightarrow$ V13	+0.0016	+0.0021	out-of-scenario hidden cells
V13 $\rightarrow$ V14a	+0.0005	+0.0004	out-of-scenario hidden cells
V14a $\rightarrow$ V14 (onset prior correction)	+0.0023	-0.0027	mined masked-layer onset events (5 flips)

The V12 miss is the reason for the protocol. Its Task 1 changes were right in the training scenario and wrong in the new ones (Section 4.2). The out-of-scenario evaluator caught this, and its predictions now match the online deltas.

3. What the generator looks like

- **Two flat plateaus.** In free flow, $v/v_f = 1.00$ at every flow level (p10–p90 within $\pm 1.5\%$). A queue sits at about $0.30\text{--}0.35 \cdot v_f$. There is no speed precursor of breakdown.
- **Queues appear as blocks.** Whole clusters of links switch on at once (for example links 4–10 of D7_I10_E). Before onset, the flow/capacity of the cluster that will queue is ordinary (0.5–0.7). The highest-flow links never queue.
- **Common-mode noise.** Free-flow speed noise has sd 0.8–1.75 km/h per panel and is nearly white in time (lag-1 autocorrelation 0.06). It is correlated about 0.3 between links 1 and 30 positions apart. The common factor has sd 0.46–1.09 km/h. Flow noise has no usable common mode.
- **Ramp sensor.** On-ramp inflow falls to 5–70% of its time-of-day profile while the mainline link it feeds is queued, and off-ramp flow rises about 10%.
- **Scenario shift.** Compared with train, per-link plateaus in validation/private move with sd about 1.3 km/h. Per-link queued-speed levels move with sd 1–6 km/h. Bottleneck clusters switch on or off; for example, D12_I5_S links 43–45 queue on 89% of train days but 45% of March days, and the onset time of D12_I5_N moves from 12.4 h to 10.8 h.

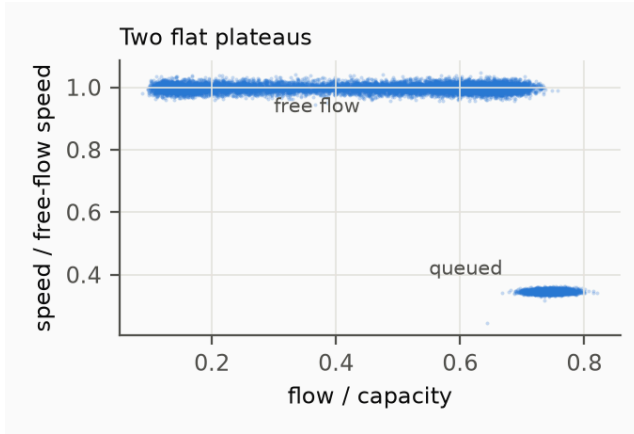


Figure 1. Speed against flow for one link of D7_I10_E (train, 15k cells): a flat free-flow branch and a flat queued plateau, no breakdown precursor.

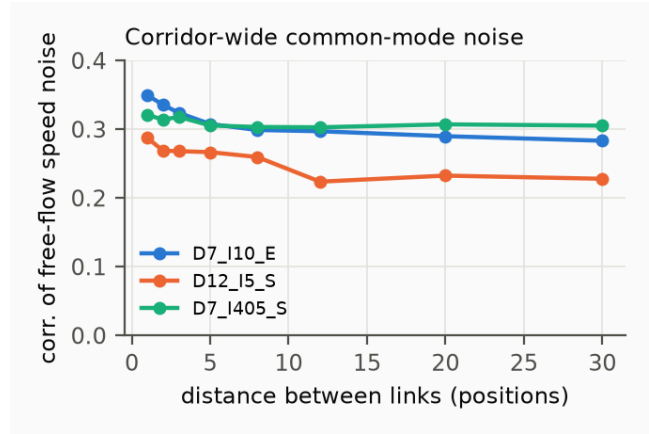


Figure 2. Correlation of free-flow speed noise between links k positions apart (train). It stays near 0.3 up to 30 positions: a corridor-wide common mode.

4. Task 1 — state reconstruction

4.1 Model

A LightGBM residual model is fitted over temporal linear interpolation: speed in km/h, and flow per lane. Features: temporal/spatial interpolations, $\pm 1/\pm 2$ neighbours, neighbour-difference estimates, time-of-day profiles, local level relative to the profile, pct_observed and link attributes. On top of these:

- **Common mode.** cf_sp is the mean of $(v - \text{plateau})$ over the free-flowing links observed at the same slot. We also use a relative version, a ± 5 -link local version and the plateau itself. Only the speed model uses them; they did not help flow.
- **Split-own statistics.** Plateaus and the gap specialist's ramp and speed/flow profiles are measured on the split's own published masked layer.
- **Gap specialist.** Validation/private blank every link for $T+5 \dots T+90$ after each Task 2 origin. Train has no such blackouts, so we place synthetic ones at onset and ongoing origins in train. A dedicated model is trained on the cells inside them, with ramp-ratio features: per-link on/off-ramp flow divided by its profile, temporal smoothing, $\pm 2/\pm 5$ -link windows, and the share of collapsed on-ramps nearby.
- **Capacity.** Out of scenario, the learning curve kept improving up to all train target cells (about 20M rows, with the matrix filled column-wise into one float32 array).

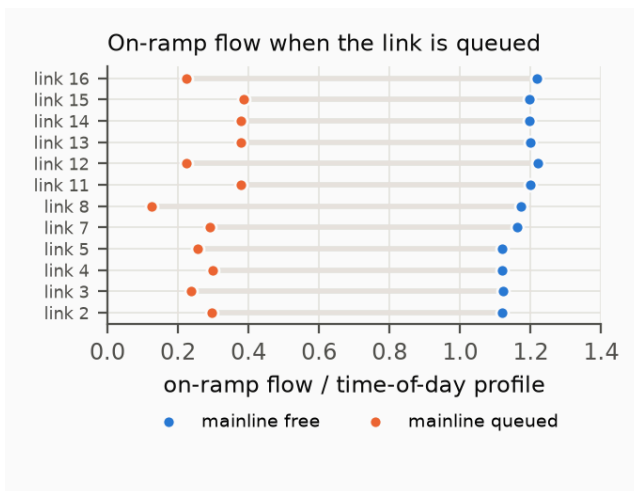


Figure 3. On-ramp flow relative to its time-of-day profile while the attached mainline link is free vs queued (D7_I405_S, train). The ramp layer stays published inside the Task 2 blackouts.

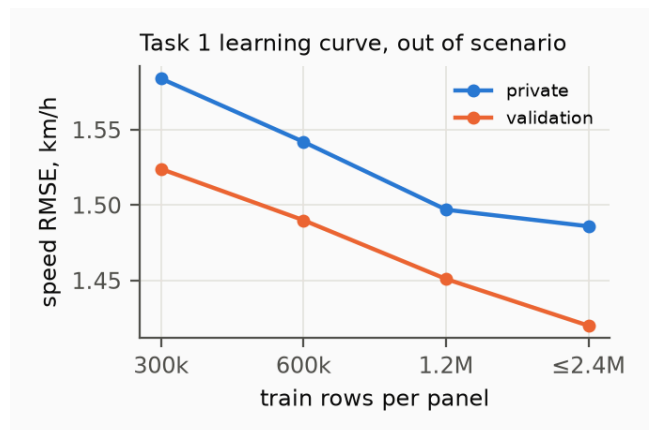


Figure 4. Task 1 speed RMSE out of scenario (hidden cells of validation/private) against the number of training rows per panel.

4.2 Results (out of scenario, hidden cells; RMSE, private / validation)

Main model	speed (km/h)	flow (veh/h/lane)
inherited model (train profiles, no common mode)	1.888 / 1.883	30.26 / 30.32
+ common mode, train plateau (V12)	1.914 / 1.892	30.26 / 30.32
+ common mode, split-own plateau (V12b)	1.584 / 1.524	30.26 / 30.32
600k rows/panel, 5000 rounds	1.542 / 1.490	30.03 / 30.03
1.2M rows/panel	1.497 / 1.451	29.82 / 29.82
all train targets (V13)	1.486 / 1.420	29.69 / 29.68

On true free-flow cells the final model reaches 1.145 km/h out of scenario. The ideal estimator "plateau + common mode" reaches 1.152, so free flow is at the noise floor. The remaining speed error comes from queue-edge cells: cells that are queued in time but not in space have RMSE 5–9.

Gap specialist (synthetic blackouts)	speed	flow
without ramps	6.32 / 6.50	68.3 / 68.9
ramps, train ramp profile (V12)	9.78 / 9.83	114 / 110
ramps, split-own ramp + speed/flow profiles (V12b, V13)	4.92 / 5.10	65.7 / 65.3
+ transductive correction (V14a)	4.50 / 4.66	60.0 / 60.0

4.3 Transductive correction (V14a)

Task 1 is offline, so the split's own published cells are legitimate training data (forum ruling 742068). We hide 6% of the observed cells of the split (6 passes), rebuild every feature and split statistic without them, and fit a correction booster (learning rate 0.03, 300 rounds) with the frozen all-data model's prediction as the initial score. The gap specialist gets the same treatment on synthetic blackouts placed on the split's own layer (200 rounds). The evaluator's hidden cells stay blank in every training frame.

Main model, out of scenario	speed	flow	panels better (val / pri)
V13 (all train targets)	1.486 / 1.420	29.69 / 29.68	
correction from the other new split, 100 rounds	1.464 / 1.413	29.66 / 29.68	
correction from the split itself, 2 passes	1.399 / 1.407	29.54 / 29.58	9 / 10
correction from the split itself, 6 passes (V14a)	1.397 / 1.400	29.50 / 29.55	9 / 10

A correction learned on the other new scenario barely helps and degrades with more rounds; the gain is scenario-specific (which links queue when, at which level), exactly what a train-only model cannot know.

5. Task 3 — physical consistency

Task 3 is computed from the Task 1 answer: $k = q/v$ and $N = kL$, and the conservation residual is taken against organizer fluxes projected onto the observations. At a target cell with observed neighbours, the residual is about $2 \cdot |N \text{ error}|$. N error is dominated by flow noise, about 2% of q , against about 1.2% for speed. A flux-based estimate cannot help either: the 5-minute flux term $dt \cdot q$ is about 20 times N , so flow noise in the fluxes swamps it. The physics score therefore follows Task 1 accuracy, mostly flow. The flow gains of the capacity steps add roughly +0.0006 through S_LWR .

6. Task 2 — queue forecasting

6.1 Onset

In every onset window the queue first appears at the last horizon step ($T+30$), on a cluster of bottleneck links. The model has three parts:

- a cluster model for which cluster activates;
- a link model inside an active cluster;
- an unconditional link model.

Decoding draws joint Monte-Carlo samples and picks the set that maximizes expected IoU. The window history ends at $T-5$. The origin-row-free refit gained $+0.034$ online.

6.2 Ongoing

The ongoing forecast is $0.5 \cdot \text{LightGBM} + 0.5 \cdot (\text{mean of CNN seeds})$ per cell and horizon step, with threshold 0.5. The CNN is a 1-D dilated residual network over links, taking 13 history steps $\times$ channels. The LightGBM uses per-cell features of the queue geometry: distance to the head and tail, run lengths, the queue's movement over the last 6 slots, up/downstream flows, and same-day pre-origin episode features. **Origin row:** the masked layer publishes row T at about 58% coverage, with only Task 1 targets blank. Training and inference both use it. Paired evaluation of the LightGBM part: in scenario $+0.0102 \pm 0.0008$, validation $+0.0231 \pm 0.0025$, private $+0.0211 \pm 0.0026$, 7/8 panels.

Ongoing IoU, out of scenario (private / validation)	
inherited ongoing (no origin row)	0.8813 / 0.8772
LightGBM with origin row	0.8863 / 0.8832
CNN with origin row (2 seeds)	0.8822 / 0.8876
blend (final)	0.8986 / 0.8980

A third CNN seed adds $+0.0002 \pm 0.0006$, so the blend is saturated.

6.3 What limits onset

When the predicted cluster is right, onset IoU is 0.94–0.98; when it is wrong, 0.14–0.16. The wrong-cluster rate is 12% in the training scenario and about 20% in the new ones. Onset IoU is 0.893 in scenario and 0.765 out of scenario. We tested every causal source of scenario evidence; none passed the adoption rule (Section 8). Suppose every other component sat at a generous ceiling simultaneously:

- S_{state} 0.954, all cells at the white-noise floor;
- S_{physics} 0.77;
- S_{ODME} 1.0;
- ongoing IoU 0.91.

Then the public prize zone (0.917) would still need onset IoU ≥ 0.872 in an unseen scenario.

A label-shift correction that did not transfer. Only 5 onset windows per corridor and split are scored, so one window moved to the right cluster is worth ≈ 0.003 . Following label-shift adaptation (Saerens et al.; Alexandari et al.) and delayed-feedback test-time adaptation, we reweighted the cluster odds by the split's earlier-day activation rate near the event time relative to train (earlier days only, compliant). On 196 mined events it flipped 5 near-ties, all correctly ($+0.012$ IoU, 2.0 SE). Online it cost 0.0027: it changed 11% of the real onset windows against 2.5% of the mined events, so the mined events did not represent the real windows. We withdrew it; onset changes now need many more flipped events before they are trusted.

7. Task 4 — OD and path-flow estimation

The link counts are noise-free. We take the L2 projection of the split's own weak prior b onto $\{A f = c, f \geq 0\}$, solved through its dual. It equals the organizers' ridge reference (NNLS, $\lambda = 0.05$) to within 0.04% L1. The relative L2 deviation from the prior reproduces five online points: L2, KL, a rescaled prior, a per-path shift and a pooled prior. L2 is optimal under it. Every alternative was worse online, by 0.11 to 0.30 S_{ODME} .

8. What did not work (out of scenario unless stated)

- **Onset physics precursors** (flat diagram, flow/capacity): none exist.
- **Onset with ramp demand**: gained on mined events from one panel, then lost 0.009 online. This was small-sample luck.
- **Onset with prior-day evidence**: cluster day-rates, activation timing with 0–4 earlier days, and same-day early evidence. None was significant, and the splits disagreed.
- **Onset calibration** (temperature, shrinkage): validation and private disagreed.
- **Ramp features for the main Task 1 model**: mixed, 12/20 panels.
- **Queued-speed plateau feature**: worse, 6/20 panels.
- **Spatiotemporal CNN imputer for Task 1** (36-slot $\times$ corridor windows, factorised dilated convolutions): it fits the training scenario's spatial patterns and does not transfer. Out of scenario its flow RMSE is about $2\times$ LightGBM's, and every blend weight was worse. The ongoing CNN with early-day channels failed the same way.
- **More ongoing CNN seeds**: saturated. Threshold 0.45: $+0.0004 \pm 0.0003$ in scenario; not adopted.

9. Submission history (public leaderboard)

Version	Change	Public
v1	interpolation, persistence, L2	0.70508
v2	static onset set	0.81819
v3–v6	LightGBM T1/T2, gap specialist, retuning, full-train T1	0.85566 $\rightarrow$ 0.86864
v9b	onset cluster mixture, LightGBM + 4-seed CNN ongoing	0.87104
V10a	origin-row-free onset refit	0.87446
V12	common mode + ramp gap (train statistics) + origin row	0.87324
V12b	split-own plateaus and profiles	0.87900
V13	Task 1 on all train targets	0.88107
V14	+ transductive Task 1 corrections + onset prior correction	0.87880
V14a	V13 + transductive Task 1 corrections	0.88147

10. Reproducibility

trafficflow/reproduce.sh is the single entry point. It rebuilds V13 and then V14a from the Kaggle release and the official toolkit (CPU only: 4 cores, 15 GB RAM, about 6 hours for V13 plus 1.5 hours for V14a). Every step caches its result and resumes after interruption. A clean verification run reproduced V13 with 14 of 174,000 Task 2 cells different, Task 4 identical and Task 1 within 0.07 km/h and 2 veh/h/lane RMSE of the submitted values, a worst-case score change of ≈ 0.003 against the 1% criterion (≈ 0.009); on the out-of-scenario hidden cells the rebuilt models score the same as the submitted ones ($\Delta \approx 0.00001$). V14a's rebuild stays within 0.12 km/h and 2.3 veh/h/lane RMSE of the submitted Task 1 values (worst case ≈ 0.004 in total). Versions are in trafficflow/requirements.txt. Code is under Apache-2.0. No external data and no pretrained models.